Emergent Gravity and a New Lagrangian: A Theory of Everything We Know

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Abstract

We consolidate a research program [58, 59, 60] into a single document with a single thesis: the gravitational sector of known physics is entanglement bookkeeping, and the unique effective Lagrangian this bookkeeping selects, once the empirically confirmed matter content is installed, is the Standard Model minimally coupled to Einstein–Cartan gravity with cosmological constant. Part I establishes the mechanism: in holographic conformal field theories the first law of entanglement, imposed on all ball-shaped regions, is equivalent to the linearized Einstein equations; the superposition principle for geometry follows as a theorem; and the extension to Riemann–Cartan geometry is derived, with torsion produced by the same bookkeeping wherever the state carries spin density. Part II assembles the Lagrangian and its certifications, including the single parameter-free prediction beyond known physics: a universal gravitational-strength axial–axial contact interaction among all fermions. The title is exact and limited: this is a theory of everything *we know*—curvature, inertia, torsion, and geometric superposition derived; gauge structure, masses, and generations installed as empirical input; finality expressly disclaimed. Appendices A and B establish the domain of validity (AdS versus de Sitter) and the theorem-grade status of Wick rotation. Appendix C answers, in numbered form, the objections a skeptical referee should raise. Every material claim is labeled **[P]** (proved/derived), **[C]** (conjectured with support), or **[O]** (open).

Epistemic Conventions

Every material claim is tagged: **[P]** proved or derived within a stated framework, with the derivation published in the primary literature cited at the point of use or presented in full here; **[C]** conjectured, supported by nontrivial consistency checks but lacking proof; **[O]** open. No claim is asserted without a tag or citation. Numerical agreement is not proof. Consensus is not proof.

Part I: Gravity from Entanglement

1 The Mechanism

1.1 Kinematics

Definition 1.1. For a spatial region A and global state $|\Psi\rangle$, the reduced state is $\rho_A = \mathrm{Tr}_{\bar{A}}\, |\Psi\rangle\langle\Psi|$, the modular Hamiltonian $H_A = -\log \rho_A$ (up to normalization), and the entanglement entropy

$S_A = -\operatorname{Tr} \rho_A \log \rho_A$.

Two exact results anchor everything. For the Rindler wedge in any Wightman QFT, the vacuum modular Hamiltonian is 2π times the boost generator (Bisognano–Wichmann [10]). For a ball B of radius R in a CFT vacuum (Casini–Huerta–Myers [33]),

$$H_B = 2\pi \int_B d^{d-1}x \, \frac{R^2 - |\vec{x} - \vec{x}_0|^2}{2R} \, T_{00}(x) \, . \tag{1}$$

Theorem 1.2 (First law of entanglement)**.** *For any family $\rho_A(\lambda)$ with $\rho_A(0) = \rho_A^{\mathrm{vac}}$,*

$$\delta S_A = \delta \langle H_A \rangle = \operatorname{Tr}\big(\delta \rho_A H_A\big) \quad \textit{at first order.} \tag{2}$$

Proof. $\delta S_A = -\operatorname{Tr}(\delta\rho_A \log \rho_A^{\mathrm{vac}}) - \operatorname{Tr}\, \delta\rho_A$; the second term vanishes by trace preservation, the first is $\delta\langle H_A\rangle$. **[P]** □

The right-hand side of (2)—the *modular energy* $E_{\mathrm{mod}}[\delta\rho] = \operatorname{Tr}(\delta\rho_A H_A)$—is the linear functional on which the entire program runs (Appendix C, Objection C.3). The exact identity behind it is $S(\rho_A \| \rho_A^{\mathrm{vac}}) = \Delta\langle H_A\rangle - \Delta S_A \geq 0$ [28, 38]: entropy's nonlinearity is quarantined in the relative entropy, which is second order and positive.

1.2 Dynamics: the equivalence theorem

Theorem 1.3 (Faulkner–Guica–Hartman–Myers–Van Raamsdonk [42]; see also [41])**.** *Let a holographic CFT$_d$ state $|\Psi\rangle = |0\rangle + \lambda|\delta\Psi\rangle$ be dual to $g = g_{\mathrm{AdS}} + \lambda h$, with δS_B computed gravitationally by Ryu–Takayanagi [26, 27] and $\delta\langle H_B\rangle$ by (1). Then $\delta S_B = \delta\langle H_B\rangle$ for all balls of all radii, centers, and frames* if and only if *$h_{\mu\nu}$ satisfies the Einstein equations linearized about AdS, sourced by $\delta\langle T_{\mu\nu}\rangle$. **[P]***

Remark 1.4 (The mechanism, stated exactly)**.** *An excitation—a quasiparticle—curves spacetime through the perturbation it induces in the* vacuum *entanglement across every partition, quantified by (2); consistency of these δS_B across all partitions forces an emergent metric obeying linearized Einstein dynamics. Intrinsic entanglement between excitations is a microstate correlation, not a source: two excitations in a Bell state versus a product state with identical stress profiles produce the identical linearized metric. **[P]** (Appendix C, Objections C.1–C.2.)*

Beyond linear order: relative-entropy identities reproduce the nonlinear Einstein equations through second order [55, 40]; Jacobson's entanglement equilibrium [47] proposes the full nonlinear equations for general backgrounds **[C]**; connectivity from entanglement [30, 37] and tensor networks [36] supply the structural picture **[P]**/**[C]** as itemized in [58].

2 Superposition as a Theorem

Proposition 2.1 (Linear order)**.** *Both sides of (2) are linear in $\delta\rho$, so the map $\delta\rho \mapsto h$ of Theorem 1.3 is linear: superposed source perturbations map to superposed metric perturbations, coherences included. Weak-field superposition of gravity is a corollary, not a postulate; naive semiclassical sourcing by $\langle T_{\mu\nu}\rangle$ of a macroscopic superposition is excluded within the framework and experimentally [16]. **[P]***

Proposition 2.2 (Macroscopic superpositions)**.** *In holographic quantum error correction the Ryu–Takayanagi area is an operator in the center of the region algebra [43, 49, 44]:* $S(\rho_A) = \mathrm{Tr}(\rho\,\widehat{\mathrm{Area}}/4G_N) + S_{\mathrm{alg}}$. *Superpositions of macroscopically distinct geometries are superpositions of area eigenstates with Born-rule statistics; the center supplies the approximate superselection by which classical geometry decoheres.* ***[P]*** *within AdS/CFT; beyond* ***[O]****. No collapse mechanism is required [20]; gravitationally mediated entanglement experiments [50, 51] discriminate empirically.* ***[C]*** *pending data.*

3 GR plus Torsion, Derived

In first-order variables (e^a, ω^{ab}; torsion $T^a = de^a + \omega^a{}_b \wedge e^b$), the Palatini action

$$S = \frac{1}{4\kappa}\int \epsilon_{abcd}\, e^a \wedge e^b \wedge R^{cd} + S_m[e, \omega, \psi]\,, \qquad \kappa = 8\pi G\,, \tag{3}$$

yields under independent variation [2, 3, 11, 25]

$$\tfrac{1}{2}\epsilon_{abcd}e^b \wedge R^{cd} = \kappa\tau_a\,, \qquad T^\lambda{}_{\mu\nu} + \delta^\lambda_\mu T^\sigma{}_{\nu\sigma} - \delta^\lambda_\nu T^\sigma{}_{\mu\sigma} = \kappa s^\lambda{}_{\mu\nu}\,. \tag{4}$$

The Cartan equation is algebraic: torsion is locked pointwise to spin density, does not propagate, and vanishes in vacuum. Eliminating it yields Riemannian GR with Belinfante source plus the Hehl–Datta contact term [6]; for Dirac matter, $\mathcal{L}_{\mathrm{HD}} = -\frac{3\kappa}{16}(\bar\psi\gamma^5\gamma^\mu\psi)(\bar\psi\gamma^5\gamma_\mu\psi)$. **[P]** The thermodynamic derivation extends to this setting [53, 57], with entropy = Area/$4G$ robust in first-order variables [46]. **[P]** within assumptions.

Proposition 3.1 (Torsionful first law, linear order)**.** *Linearizing about a maximally symmetric vacuum: background torsion vanishes (Lorentz invariance + algebraicity); the system decomposes into a metric sector obeying linearized Einstein equations with Belinfante source and an algebraic torsion sector fixed by* $\delta\langle s^\lambda{}_{\mu\nu}\rangle$*; Hehl–Datta is quadratic and drops. Since the CHM Hamiltonian* (1) *is built from the Belinfante tensor, Theorem 1.3 extends verbatim: the first law is equivalent to linearized Einstein–Cartan gravity, with torsion produced by the same bookkeeping wherever the state carries spin.* ***[P]***

The companion note [59] resolves the case of explicit spin-current sources: for minimal Dirac matter only the axial (totally antisymmetric) source couples, pure-gauge sources act only through their curl, the torsionful first law holds unconditionally on null planes (via the Markov property [54] and the FLPW deformation theory [48, 56]) and conditionally for finite balls; mixed-symmetry sources through non-minimal matter are relevant deformations that destroy the construction. **[P]**/**[O]** as itemized there. This *modular selection rule* is what Part II uses.

Part II: The Lagrangian

4 The New Standard Model

Definition 4.1. The New Standard Model (NSM) is the action (5)—the full $SU(3)_c \times SU(2)_L \times U(1)_Y$ Standard Model [1, 4, 5], three generations, Higgs sector [34, 35], neutrino masses [22] (Dirac or Weinberg-operator, empirically open **[O]**), minimally coupled through vielbein and independent spin connection to Einstein–Cartan gravity with cosmological constant—*together with* the certifications of Section 5.

$$\boxed{S_{\mathrm{NSM}} = \frac{1}{4\kappa}\int \epsilon_{abcd}\Big(e^a\wedge e^b\wedge R^{cd} - \frac{\Lambda}{6}\,e^a\wedge e^b\wedge e^c\wedge e^d\Big) + \int d^4x\, e\,\mathcal{L}_{\mathrm{SM}}[e,\omega,\psi,A,H]} \tag{5}$$

with $\mathcal{L}_{\mathrm{SM}}$ the standard gauge, fermion, Higgs, and Yukawa terms, fermions covariantized with the full connection $D_\mu = \partial_\mu + \frac{1}{4}\omega_{ab\mu}\gamma^{ab} + ig\cdot A_\mu$. Gauge field strengths use the exterior derivative alone (gauge invariance forbids torsion coupling [11, 24]); scalars carry no spin. *Only fermions talk to torsion.* **[P]**

Eliminating the algebraic torsion yields the single interaction beyond the SM and Einstein gravity:

$$\boxed{\mathcal{L}_{\mathrm{HD}} = -\frac{3\kappa}{16}\,\mathcal{J}_5^\mu\mathcal{J}_{5\mu}\,, \qquad \mathcal{J}_5^\mu = \sum_{f\in\text{all fermions}} \bar\psi_f\gamma^5\gamma^\mu\psi_f} \tag{6}$$

—universal, flavor-blind, cross-species, gravitational strength, with *no free parameter*: the coefficient is fixed by G. An output, not an input. **[P]**

5 The Certifications

Theorem 5.1 (Gravity certified)**.** *At linear order the metric dynamics of* (5) *is equivalent to the entanglement first law on all balls (Theorem 1.3, Proposition 3.1): curvature and inertia are the unique consistent response to perturbations of vacuum entanglement.* ***[P]*** *within the holographic regime; nonlinear torsionful equations under Clausius assumptions [18, 53].*

Theorem 5.2 (Torsion structure certified)**.** *The minimal axial torsion coupling of* (5) *is the unique structure surviving the modular selection rule of [59]: mixed-symmetry couplings through non-minimal matter are relevant deformations destroying the modular framework.* ***[P]*** *at the level of the dimensional and symmetry argument.*

Theorem 5.3 (Superposition certified)**.** *Section 2 applies verbatim: the NSM requires no metric quantization as an independent postulate and no collapse mechanism.* ***[P]*** *within AdS/CFT;* ***[O]*** *beyond.*

Remark 5.4. *No analogous certification exists for the gauge and Yukawa structure: the matter content is empirical input, and whether gauge dynamics, chirality, and three generations admit an entanglement-theoretic derivation is the deepest question this program can pose.* ***[O]***

6 Falsifiable Consequences; Declared Limits

Consequences: (1) the parameter-free contact term (6), consistent with torsion bounds [29, 24], undetectable at collider energies **[P]**; (2) torsion-induced bounce replacing the initial singularity [8, 31] **[C]**; (3) gravitationally mediated entanglement as a live falsifier of the gravitational sector [50, 51] **[C]**; (4) infrared deviations at the Milgrom scale if the de Sitter extension follows the emergent route [52, 17] **[C]**.

Declared limits: the NSM does not unify gauge couplings, explain masses or generations, resolve strong CP, supply a dark-matter particle, determine Λ, or provide its own UV completion—the last by construction, since the Lagrangian is asserted to be emergent bookkeeping for a deeper description **[O]**. *A theory of everything we know is not a theory of everything*; on this program's epistemology, finality is not claimable.

A Domain of Validity: AdS versus de Sitter

A.1 Why the proofs live in AdS

The **[P]**-grade chain requires three ingredients AdS supplies and de Sitter does not: a timelike boundary carrying a complete unitary CFT with the Maldacena dictionary [21] (de Sitter's candidate dS/CFT [23] is Euclidean, generically non-unitary, and **[O]**); a global timelike Killing structure guaranteeing the Bisognano–Wichmann machinery (de Sitter has only observer-dependent static patches with Gibbons–Hawking horizon entropy [13], microscopic description **[O]**); and theorem-grade entanglement technology [26, 40, 45], unavailable at that grade in de Sitter **[O]**. Rule: certify in AdS, extrapolate with labels.

A.2 Why the mechanism governs our space

Proposition A.1 (Flat-space inputs)**.** *The quantum-informational inputs—CHM* (1)*, the first law, relative-entropy positivity and monotonicity, the null-plane Markov property [54]—are theorems of QFT on Minkowski space, independent of holography. The vacuum of the Standard-Model fields in our universe carries the structure the mechanism runs on.* ***[P]***

Proposition A.2 (Locality)**.** *The thermodynamic route [18, 53] uses only local Rindler horizons, available through every point of every spacetime, and yields the field equations pointwise; AdS/CFT upgrades the local inputs to theorems.* ***[P]*** *structure; general-background transfer [47]* ***[C]****.*

Proposition A.3 (Infrared decoupling of Λ)**.** *With $\Lambda \simeq 1.1 \times 10^{-52}\,\mathrm{m}^{-2}$, the local relevance measure ΛL^2 is $\sim 10^{-52}$ (laboratory), $\sim 10^{-30}$ (1 AU), $\sim 10^{-11}$ (10 kpc), $\mathcal{O}(1)$ only at the Hubble radius. Below the Hubble scale the AdS/dS/flat distinction is invisible to local experiment; the certified regime is $\Lambda L^2 \ll 1$.* ***[P]***

Proposition A.4 (Where the difference bites)**.** *The sign of Λ is structural only near the horizon scale, where de Sitter has an observer horizon with entropy [13] and possibly a volume-law entanglement contribution [52]. Any modification appears at accelerations $\sim cH_0$; Milgrom's $a_0 \simeq 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}} \approx cH_0/6$ [17] sits exactly there. Coincidence* ***[P]****; explanation* ***[C]****; microtheory* ***[O]****. The proof's edge coincides with observation's anomalies.*

B The Status of Wick Rotation

Euclidean continuation enters at three load-bearing points, each covered by a theorem. **Reconstruction:** Osterwalder–Schrader [7] uniquely reconstructs a unitary, causal, positive-energy Lorentzian QFT from reflection-positive Euclidean data; the boundary theories used here satisfy the hypotheses. **[P]** **Instantiation:** Bisognano–Wichmann [10] *is* the rotation, proved once at the foundational level: wedge modular flow is the boost, with (1) its conformal image;

Tomita–Takesaki theory [19] performs no rotation at all; the Unruh temperature [12] is the KMS countersignature. **[P] The fragile uses are absent from [P] claims:** (i) the replica limit $n \to 1$ [39] is never performed at linear order—the first law is an operator identity and Theorem 1.3 uses RT as an independently established input; (ii) the Euclidean gravitational path integral, unbounded below [14], is never performed—gravity is emergent and every gravitational statement is boundary entanglement; (iii) de Sitter Euclidean vacua are genuinely subtle and lie entirely inside the **[O]** region of Appendix A—the two frontiers coincide, a consistency check on the labeling. **[P]** Fermion γ^5 conventions touch one certified claim (the anomaly caveat in [59]) and are fixed convention-independently by the index theorem [15]. **[P]**

C Anticipated Objections, Answered

Each objection below was raised, in some form, during the development of this program. Answers cite the theorem that settles the point or the tag that concedes it.

Objection C.1 ("This is Verlinde's entropic gravity, and/or it runs on 'relative entropy.'")**.** Neither. The entropy in Theorems 1.2–1.3 is *entanglement entropy across a partition*—relationally *defined* (no partition, no entropy) but not "relational entropy" as a term of art, and not Verlinde's coarse-grained screen entropy [32], which requires $N \gg 1$ bits at finite temperature and is undefined for a single excitation. Relative entropy enters in one exact role: as the positive second-order remainder $S(\rho\|\rho_{\mathrm{vac}}) = \Delta\langle H\rangle - \Delta S$ [28, 38], whose positivity delivers energy conditions and second-order dynamics [55]. Verlinde's 2016 volume-law proposal [52] is a **[C]** extension confined to Appendix A's frontier, never load-bearing for a **[P]**. **[P]**

Objection C.2 ("So entangled particles gravitate toward each other because they're entangled?")**.** No, and the theorem's content forbids it. Curvature at linear order is fixed entirely by $\delta\langle T_{\mu\nu}\rangle$ (Remark 1.4): a Bell pair and a product pair with identical stress profiles produce identical metrics. The gravitating object is the excitation's perturbation of *vacuum* entanglement, ball by ball; pair entanglement is a microstate correlation invisible to single-region reduced states. Whether ER=EPR [37] endows pairs with a Planckian geometric dual is **[C]** and carries no classical curvature signature in any case. **[P]**

Objection C.3 ("A thermodynamic origin for gravity forfeits quantum superposition. What is even linear here?")**.** The linear functional is the *modular energy* $E_{\mathrm{mod}}[\delta\rho] = \mathrm{Tr}(\delta\rho_A H_A)$, and the first law says δS_A *equals* it: entropy's nonlinearity is entirely quarantined in the positive second-order relative entropy. Both sides of the equivalence in Theorem 1.3 are therefore linear in $\delta\rho$, making weak-field superposition a corollary (Section 2); macroscopic superpositions of distinct geometries are area-operator superpositions [49]. The naive semiclassical alternative is excluded experimentally [16]; the collapse alternative [20] is empirically discriminable [50, 51]. **[P]** within stated frameworks.

Objection C.4 ("Torsion is ad hoc, dead since the 1970s, and unobservable.")**.** Torsion is not added; it is what the first-order formalism *already contains* the moment fermions exist: spinors couple to the connection, the connection's variation is the Cartan equation (4), and refusing torsion is the ad hoc step. It is algebraic (no new propagating fields), experimentally bounded and consistent [29, 24], selected uniquely by the modular argument [59], and its one consequence (6) is parameter-free. "Unobservable at colliders" is conceded **[P]** and is exactly what a gravitational-strength contact term must be; the high-density regime [8, 31] is where it earns its keep. **[C]** there.

Objection C.5 ("Why is everything proved in anti-de Sitter when the universe is de Sitter?")**.** Because in AdS the claim is a theorem and in de Sitter it is a hope, and the discipline of this document is never to blur that line. The full argument is Appendix A: AdS supplies the boundary, the dictionary, and the theorem-grade technology; the mechanism so certified is *local*, built from flat-space theorems our fields satisfy; the measured Λ decouples below the Hubble scale ($\Lambda L^2 \ll 1$); and the one regime where the certification runs out is the one regime where observation shows anomalies. AdS is the wind tunnel, not the sky. **[P]**/**[C]**/**[O]** as tagged there.

Objection C.6 ("Euclidean methods smuggle in unjustified analytic continuation.")**.** Appendix B: the rotation used is licensed by Osterwalder–Schrader, instantiated by Bisognano–Wichmann, countersigned by the Unruh/KMS thermality, and the three genuinely fragile continuations—replica limits, the Euclidean gravitational path integral, de Sitter vacua—are respectively never used at linear order, structurally bypassed by emergence, and coincident with the declared **[O]** frontier. **[P]**

Objection C.7 ("You have a Lagrangian for QFT plus torsionful gravity—so you claim a fundamental theory.")**.** Two levels, kept separate. At the effective level, yes: (5)–(6) is the complete EFT of known physics, quantizable below M_{Pl}, and *certified* in its gravitational and torsion structure. At the fundamental level, no, by construction: in this framework the Lagrangian is emergent bookkeeping; the microscopic description is the boundary theory of a holographic pair, and no first-principles selection of that pair exists **[O]**. The sharpest open prize is whether the Hehl–Datta coefficient itself can be derived from relative-entropy positivity at joint second order, which would make the entire EFT an entanglement theorem. **[O]**

Objection C.8 ("The title claims a theory of everything.")**.** Read it in full: a theory of everything *we know*. Derived: curvature, inertia, torsion, geometric superposition, and one new interaction with no free parameter. Installed as empirical input: gauge group, representations, Yukawas, generations, Λ. Declared open: their origin, the UV completion, de Sitter. The program's own epistemology—emergence without a final layer—forbids the stronger reading, and Section 7 enumerates the exclusions with the same rigor as the claims. A referee who finds the title too strong is invited to find a single sentence in the body that is. **[P]** (as a statement about the document).

Certified Summary

Gravity is entanglement bookkeeping **[P]**; its torsionful extension is derived and modular-selected **[P]**; geometry superposes as a theorem **[P]**; the unique resulting effective Lagrangian of known physics is (5) with the single parameter-free consequence (6) **[P]**; the domain of validity and the analytic-continuation machinery are theorem-anchored with their frontiers labeled **[P]**/**[O]**; and everything not derived is named. This is the current certified layer—the next turtle, load-bearing and inspected.

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